

Ice Task 4

Theories:

1. Secure software development lifecycle

Secure software development lifecycle (secure SDLC) is a framework that is widely recognized which integrates the security practices into all the phase of the traditional software development lifecycle (SDLC) and unlike the conventional approaches where security is firstly addressed during the testing phase. By embedding security from the initial planning and design phases/stages through to the other phases/stages (implementation, testing, deployment and maintenance), secure SDLC promotes a proactive approach. (OWASP, 2023)

According to OWASP (2023) secure SDLC involves key practices such as secure coding, code reviews, threat modelling and frequent security testing throughout the whole development process. The NIST (2022) emphasizes that by integrating security in the early stages of the lifecycle helps to reduce vulnerabilities and enhance the overall resilience of software systems.

How the theory applies to my proposed study

The secure SDLC framework provides a structured approach for analysing how security practices can be integrated into the development of the banking web-based applications in South Africa. Since these applications deal with sensitive and confidential financial information by applying secure SDLC principles this enables early detection of vulnerabilities and reduces the likelihood of cyberattacks. (NIST, 2022)

Justification

This is highly relevant because it directly addresses the research problem, it provides a theoretical and practical foundation for evaluating how security may be enhanced in banking applications. (OWASP, 2023)

References

NIST, 2022. Secure Software Development Framework (SSDF) Version 1.1: Recommendations for Mitigating the Risk of Software Vulnerabilities. [online] Available at: <
<https://csrc.nist.gov/pubs/sp/800/218/final> > [Accessed 21 March 2026].

OWASP, 2023. Secure development and integration. [online] Available at: <
<https://devguide.owasp.org/en/02-foundations/02-secure-development/> > [Accessed 21 March 2026].